

3. Huntington's disease (HD) is an inherited disease that causes nerve cells in the brain to degenerate. People are born with the defective gene, but symptoms usually don't appear until middle age. Early symptoms of HD may include uncontrolled movements, clumsiness or balance problems. The disease is caused by an inherited defect in a single gene. With only one copy of the defective dominant allele (**H**), a person can develop the disease.

(a) State the meaning of the following terms:

[3]

(i) allele

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(ii) dominant allele

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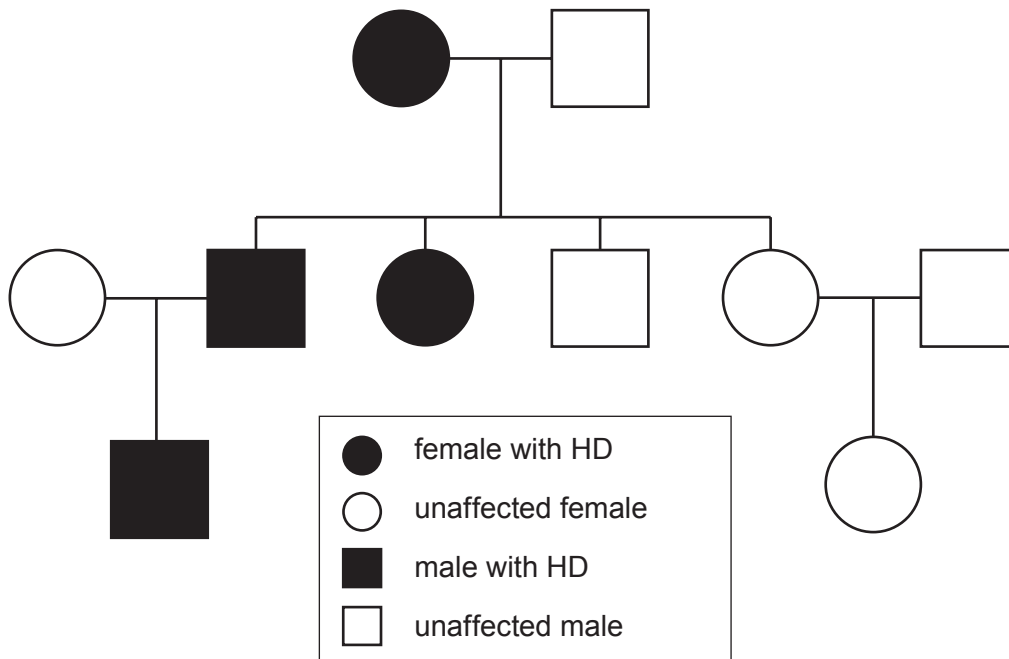
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(iii) genotype

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- (b) The diagram shows a family tree identifying those who are unaffected (do not have HD) and those with HD.



Draw **two** Punnett squares and **compare** the percentage chances of a child developing HD if:

- one parent is unaffected and one parent is heterozygous
- both parents are heterozygous

[5]

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- (c) Discuss the ethical problems which may arise when an individual undergoes genetic screening and finds out they have the allele for HD. [2]

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